Unit – II
Cloud Computing Architecture & Deployment Model

2.1.1 Introduction Cloud Computing Architecture

- Cloud Computing Architecture is a combination of components required for a Cloud Computing service.
- A Cloud computing architecture consists of several components like a frontend platform, a backend platform or servers, a network or Internet service, and a cloud-based delivery service.
- The cloud architecture is divided into 2 parts

1. Frontend:
   - It is a GUI (Graphical User Interface) which is use by user side.
   - The front end consists of the client part of a cloud computing system.
   - It comprises interfaces and applications that are required to access the Cloud computing or Cloud programming platform.
   - The front end includes web servers (Chrome, Firefox, Opera, etc.), clients, and mobile devices.

2. Backend :
   - The back end refers to the cloud itself, it provides the resources required for cloud computing services.
   - It consists of virtual machines, servers, data storage, security mechanisms, traffic control mechanisms, etc.
   - It is under the provider's control.

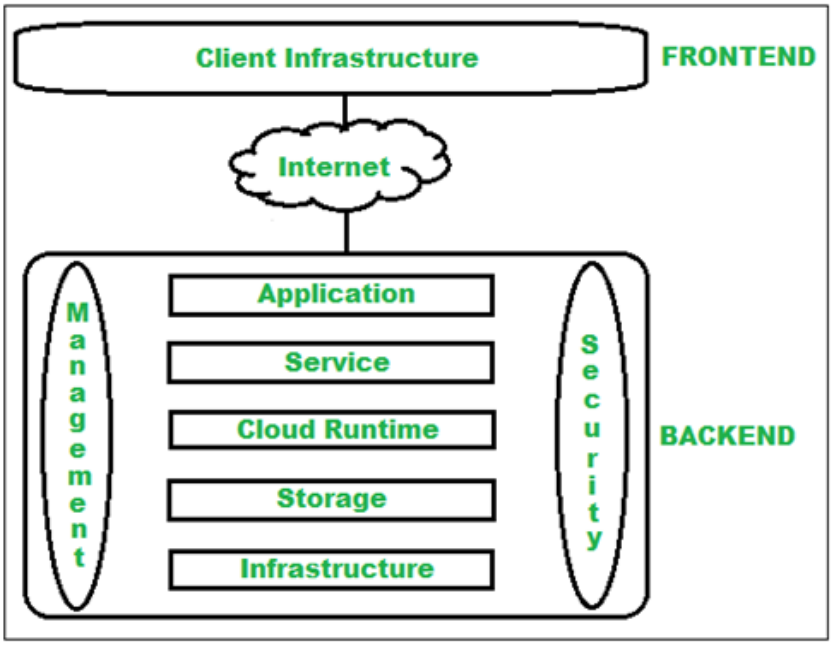


Fig. Cloud Computing Architecture

2.1.2 Components of Cloud Computing Architecture

- **Client Infrastructure**
- **Cloud access device**
- **Application**
- **Services**
- **Runtime cloud**
- **Storage**
- **Infrastructure**
- **Security**

1. Client Infrastructure

- Client Infrastructure is a Front end component.
- It provides GUI (Graphical User Interface) to interact with the cloud.

2. Cloud access device

- Cloud access devices enables users to access their files and data from the cloud, providing convenience and flexibility.
- Smartphone, Tablet, Mobile… devices can connect to cloud storage services through dedicated applications.

3. Application

- Application in backend refers to a software or platform to which client accesses.
- Means it provides the service in backend as per the client requirement.

4. Services

- The service component manages which type of service you can access according to the client's requirements.
- Three Cloud computing services are:
  - Software as a Service (SaaS)
  - Platform as a Service (PaaS)
  - Infrastructure as a Service (IaaS)

5. Runtime cloud

- Runtime cloud in backend provides the execution and Runtime platform/environment to the Virtual machine.

6. Storage

- Storage is another important Cloud computing architecture component.

- It provides a large amount of storage capacity in the Cloud to store and manage data.

7. Infrastructure

- It offers services on the host level, network level, and application level.
- Cloud infrastructure includes hardware and software components like servers, storage, network devices, virtualization software, and various other storage resources that are needed to support the cloud computing model.

8. Security

- Security is an in-built back end component of cloud computing.
- It implements a security mechanism in the back end.
- It secures cloud resources, systems, files, and infrastructure to end-users in the backend.

9. Management.

- It is use to manage all the components in back-end.

2.2.1 Types of Cloud Deployment Model

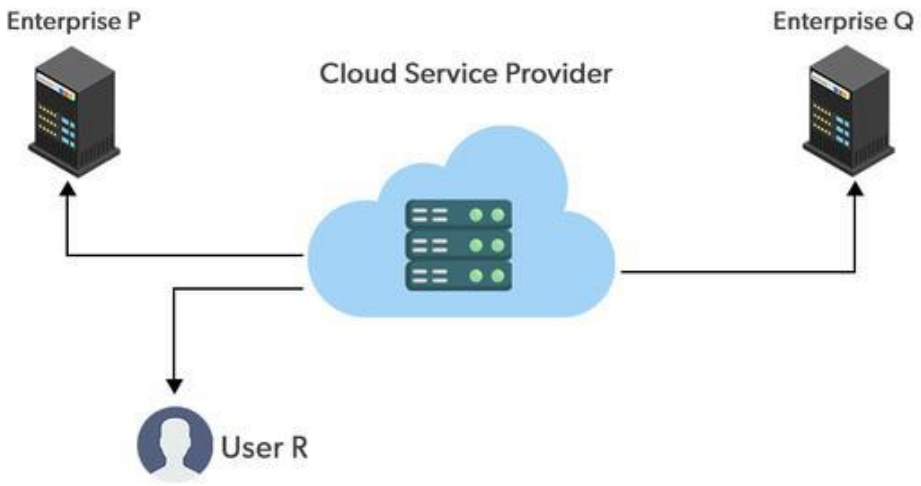
- Deployment is movement of software, hardware ,computing resources, processing power to remote servers on internet.
- Cloud Deployment Model acts as a virtual computing environment that offers a choice of deployment model according to how much data users want to store and who will have access to the infrastructure.
- It signifies how servers are deployed and provisioned over the internet so that various organizations and companies can access these servers without configuring them.
- Different types of cloud computing deployment models are as below.

1) Public cloud
2) Private cloud
3) Hybrid cloud
4) Community cloud

Public cloud:

- Public cloud As the name suggests, this type of cloud deployment model supports all users who want to make use of a computing resource, such as hardware (OS, CPU, memory, storage) or software (application server, database) on a subscription basis.
- The public cloud makes it possible for anybody to access systems and services.
- The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups.
- The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer.

- It is a type of cloud hosting that allows customers and users to easily access systems and services.
- This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers.
- In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis. For example, Google App Engine etc.



Advantages of the Public Cloud Model

- Minimal Investment: Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- No setup cost: The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- Infrastructure Management is not required: Using the public cloud does not necessitate infrastructure management.
- No maintenance: The maintenance work is done by the service provider (not users).
- Dynamic Scalability: To fulfill your company's needs, on-demand resources are accessible.

Disadvantages of the Public Cloud Model

- Less secure: Public cloud is less secure as resources are public so there is no guarantee of high-level security.
- Low customization: It is accessed by many public so it can't be customized according to personal requirements.

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Private cloud:

- The private cloud deployment model is the exact opposite of the public cloud deployment model.
- Private cloud is infrastructure used by a single organization. There is no need to share your hardware with anyone else
- Such infrastructure may be managed by the organization itself to support various user groups, or it could be managed by a service provider that takes care of it either on-site or off-site.
- The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department.
- Private clouds are more expensive than public clouds due to the capital expenditure involved in acquiring and maintaining them.
- However, private clouds are better able to address the security and privacy concerns of organizations today.

Advantages of the Private Cloud Model

- Better Control: You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- Data Security and Privacy: It's suitable for storing corporate information to which only authorized staff has access
- Supports Legacy Systems: This approach is designed to work with legacy systems that are unable to access the public cloud.
- Customization: Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

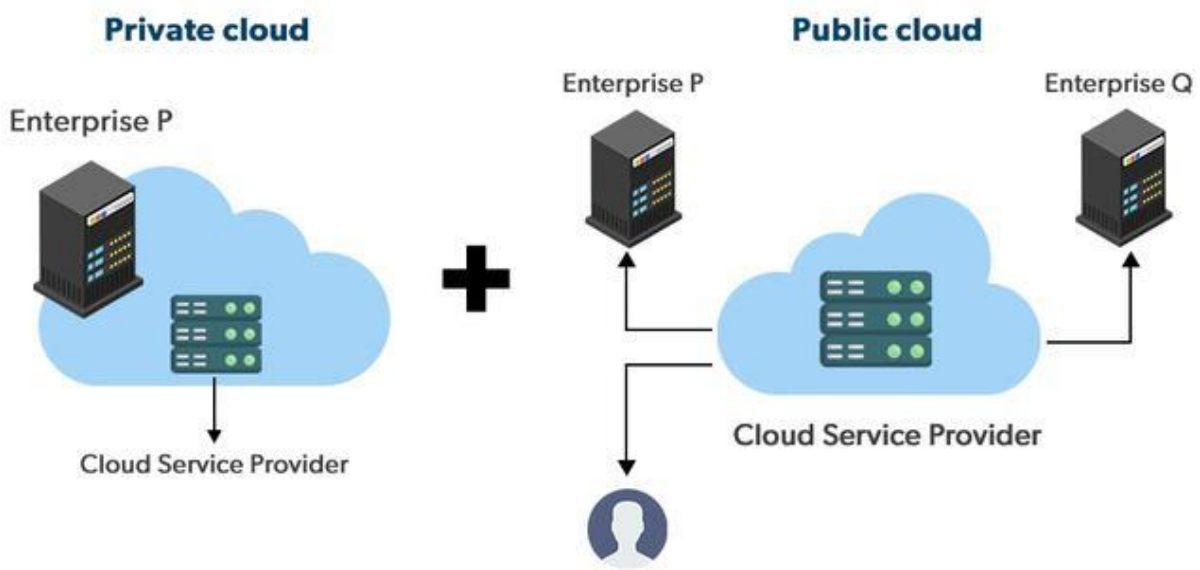
Disadvantages of the Private Cloud Model

- Less scalable: Private clouds are scaled within a certain range as there is less number of clients.

- Costly: Private clouds are more costly as they provide personalized facilities.
- High Maintenance - Since it is managed in-house, the maintenance costs also increase.

Hybrid cloud:

- In a hybrid cloud, an organization makes use of interconnected private and public cloud infrastructure.
- Both the private and public cloud work together to meet our organization requirements.
- It offers the best of both the worlds
- Many organizations make use of this model when they need to scale up their IT infrastructure rapidly, such as when leveraging public clouds to supplement the capacity available within a private cloud.
- For example, if an online retailer needs more computing resources to run its Web applications during the holiday season it may attain those resources via public clouds.
- With a hybrid solution, you may host the app in a sesure environment via Provate cloud and also taking advantage of the public cloud's cost savings.
- Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.
- Hybride cloud = Private cloud + Public cloud



Advantages of the Hybrid Cloud Model

- Flexibility and control: Businesses with more flexibility can design personalized solutions that meet their particular needs.
- Cost: Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it. As compare to private cloud costing is low.

- Security: Because data is properly separated, the chances of data theft by attackers are considerably reduced.
- Scalibility : We can up/down resources as per our requirments.

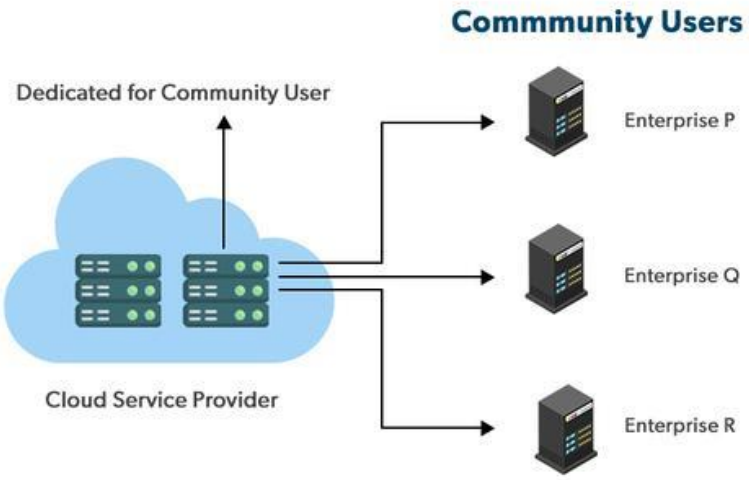
Disadvantages of the Hybrid Cloud Model

- Difficult to manage: Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- Slow data transmission: Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

Community cloud:

- This deployment model allows systems and services to be accessible by a group of organizations.
- It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business.
- The infrastructure of the community could be shared between the organization which has shared concerns or tasks.
- It is generally managed by a third party or by the combination of one or more organizations in the community.

- Example: Universities cooperating in certain areas of research, or police departments within a county or state sharing computing resources. Access to a community cloud environment is typically restricted to the members of the community.



Advantages of the Community Cloud Model

- Cost Effective: It is cost-effective because the cloud is shared by multiple organizations or communities.

- Security: Community cloud provides better security.
- Shared resources: It allows you to share resources, infrastructure, etc. with multiple organizations.
- Collaboration and data sharing: It is suitable for both collaboration and data sharing.

Disadvantages of the Community Cloud Model

- Limited Scalability: Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
- Rigid in customization: As the data and resources are shared among different organizations according to their mutual interests if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.

2.3.1 Key drivers to adopting the cloud

We move from on-site storage to cloud based storage because of following key features.

1. Cost Savings

- On-premise solutions are often quite expensive, with various costs adding up.
- These include necessary hardware and implementation costs along with costs related to continuous management and updating required from in-house personnel.
- Moving to the cloud, on the contrary, gives organizations the benefit of eliminating the high cost of hardware and installation as it allows them to opt for a subscription-based model that suits their budget.
- Most cloud solutions provide enterprises with the ability to pay for exactly what they need instead of paying for unwanted or unused services.

2.Security

- User data is more secure on cloud as compare to physical device.
- The service providers ensure along with all the new and exciting features entail an update in security through audit and compliance.
- To further enhance clients satisfactory, all activities on the cloud will be closely monitored and frequently audited by a third party to ensure that security standards are met.
- Cloud computing actually makes servers and storage more secure by defining different levels of security by implementing robust intrusion detection and prevention systems, firewalls and performing security audits regularly in their data centers to protect data.

3. Flexibility

- The nature of cloud computing is such that it allows organizations to allow services more quickly and as per company needs.
- This is because cloud computing helps in distributing workloads across the company and can be remotely accessed by end-users, irrespective of their location.

- Companies could even hire a global and, perhaps, cheaper workforce when they use cloud computing.

4. Mobility

- Mobility means no dependency on device or location.
- User can work at any time, any place , on any device.
- Cloud computing allows mobile access to corporate data via smartphones and devices like PC, Tablet.
- Staff with busy schedules, or who live a long way away from the corporate office, can use this feature to keep instantly up to date with clients and co-worker.
- Through the cloud, you can offer conveniently accessible information to staff who travel or remote employees, for better work-life balance.

5. Insight

- Many cloud-based storage solutions offer integrated cloud analytics for a bird's-eye view of your data.
- With your information stored in the cloud, you can easily implement tracking mechanisms and build customised reports to analyse information organisation wide.
- From those insights, you can increase efficiencies and build action plans to meet organisational goals.

6. Increased Collaboration

- If your business has less employees, then you should be making collaboration with other organization.
- After all, there isn't much point to having a team if it is unable to work like a team.
- Team members can view and share information easily and securely across a cloud-based platform.
- Some cloud-based services even provide collaborative social spaces to connect employees across your organization, therefore increasing interest and engagement

7.Quality Control

- In a cloud-based system, all documents are stored in one place and in a single format.
- With everyone accessing the same information, you can maintain consistency in data, avoid human error, and have a clear record of any revisions or updates.
- There are few things as detrimental to the success of a business as poor quality and inconsistent reporting.

8. Disaster Recovery

- When data is stored offline, if fire or flood damage physical device than there is no recovery or backup for that data.
- Damage to infrastructure also leads to data loss.
- Downtime in your services leads to lost productivity, revenue, and brand reputation.
- When we use cloud base storage , data recovery is easy.
- Most of cloud users claim disaster recovery in few hours.

9. Loss Prevention
   - If your local hardware generate a problem or stolen, you might end up permanently losing your data.
   - This is a more common problem than you might realize computers can malfunction for many reasons
   - If you aren't on the cloud, you're at risk of losing all the information you had saved locally.
   - With a cloud-based server, however, all the information you've uploaded to the cloud remains safe and easily accessible from any computer with an internet connection, even if the computer you regularly use isn't working.

10. Automatic Software Updates
   - Cloud-based applications automatically refresh and update themselves, instead of forcing an IT department to perform a manual organization wide update.
   - For those who have a lot to get done, there isn't anything more irritating than having to wait for system updates to be installed.
   - This saves valuable time and money spent on outside IT consultation.

11. Competitive Edge
   - While cloud computing is increasing in popularity, there are still those who prefer to keep everything local.
   - That's their choice, but doing so places them at a distinct disadvantage when competing with those who have the benefits of the cloud at their fingertips.
   - If you implement a cloud-based solution before your competitors, you'll be further along with higher outcome.

12. Sustainability
   - On-premise data centre consume more energy and increase the emission of harmful gases like carbon dioxide.
   - Moving to cloud computing help companies cut down their energy requirements and achieve sustainable growth.
   - Another important objective of green computing is facilitating remote and hybrid work culture.
   - Remote work allows companies to use smaller office spaces and reduces the infrastructure needs of employees. Similarly, employees do not need to commute to the office, which reduces the use of vehicles and hence cuts down on the emission of greenhouse gases.

2.3.2 The impact of Cloud computing on Users

- Cloud Computing allows users to access and store data on remote servers, offering flexibility, scalability, and cost-efficiency.
- This technology eliminates the need for local servers or personal devices to handle applications, including data storage, data processing and data security.
- Below are some of the critical areas of our daily lives that have seen a revolution by cloud computing.

Education

- In recent days, educational institutions have been adopting cloud computing because it offers a wide range of benefits to the students, staff, and institutions.
- Cloud computing has enabled students to access data anywhere and at any time.
- Students can enrol online and participate in online learning activities.
- Cloud computing has enabled institutions to use the storage cloud to store large amounts of data securely without installing a complicated and expensive infrastructure.
- For students in less developed countries, cloud computing technology has enabled them access to educational websites and cloud knowledge-sharing forums and communities.

Social impact

- The big industries such as music and video live stream, online gaming, and google search engine is living proof of how much cloud computing has greatly impacted our lives.
- An event in one part of the world can reach all the corners of the world.
- We are able to live stream football matches or watch our favourite movies on Netflix because of cloud computing.
- Looking for long lost high school friends has never been easy, but with the development of social networks like Facebook and Instagram, it's much easy to connect with friends and people through the online community.

Health

- Cloud computing technology is gaining pace in the healthcare industry.
- Its application includes managing and sharing patient data to reduce operational costs, such as data storage devices.
- In addition, patients are able to access their medical records through pervasive devices like mobile phones.
- This has seen the medical sector move from the traditional intranet to a quick and secure solution through cloud computing.

Navigation

- Most modern vehicles come with preinstalled navigation programs.
- Cloud computing has enabled these navigation programs to store large amounts of data and regularly update it to assist you.
- Whenever you use these tools, you are indirectly utilizing the cloud to allow you to move from one point to another.

Online shopping

- Online shopping grows day by day.
- Most online shopping platforms are using the cloud to store important client information to provide better customer experience and come up with a better marketing strategy.

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- They use the analytical data provided like, recently view or provide customer recommendations as a result of cloud computing.

Personal storage

- If you ever want to send a large file to your friend, chances are you will use Dropbox.
- If you have a personal Gmail or Outlook email service, all your personal data are stored on Google (News - Alert)'s or Microsoft's cloud server. This is possible because of cloud computing.

2.3.3 A Comparative Analysis of Cloud Deployment Models

Important Factors to Consider	Public	Private	Hybrid	Community
Setup and ease of use	Easy	Requires professional IT Team	Requires professional IT Team	Requires professional IT Team
Data Security and Privacy	Low	High	High	Very High
Scalability and flexibility	High	High	High	Fixed requirements
Cost Effectiveness	Most affordable	Most expensive	Cheaper than private but more expensive than public	Cost is distributed among members
Reliability	Low	High	High	Relatively Higher
Data Control	Low as the service provider exercises complete control over data	Very high as the user has ownership	Very high if the environment is right	High, provided there is collaboration among the members
Demand for in-house hardware	No	Yes	No need for in-house hardware	No

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